

Feel free to work with other students, but make sure you write up the homework and code on your own (no copying homework *or* code; no pair programming). Feel free to ask students or instructors for help debugging code or whatever else, though. The starter code for problem 2 part c and d can be found under the Resource tab on course website.

Note: You need to create a Github account for submission of the coding part of the homework. Please create a repository on Github to hold all your code and include your Github account username as part of the answer to problem 2.

1 (Linear Transformation) Let $\mathbf{y} = A\mathbf{x} + \mathbf{b}$ be a random vector. show that expectation is linear:

$$\mathbb{E}[\mathbf{y}] = \mathbb{E}[A\mathbf{x} + \mathbf{b}] = A\mathbb{E}[\mathbf{x}] + \mathbf{b}.$$

Also show that

$$\text{cov}[\mathbf{y}] = \text{cov}[A\mathbf{x} + \mathbf{b}] = A\text{cov}[\mathbf{x}]A^\top = A\mathbf{\Sigma}A^\top.$$

We can write the expectation as

$$\mathbb{E}[\mathbf{x}] = \int \mathbf{x} * f(\mathbf{x})d\mathbf{x}$$

where $f(\mathbf{x})$ is the probability density function. So we can rewrite our function as:

$$\mathbb{E}[\mathbf{y}] = \mathbb{E}[A\mathbf{x} + \mathbf{b}] = \int (A\mathbf{x} + \mathbf{b}) * f(\mathbf{x})d\mathbf{x}$$

When fully expanded our function becomes:

$$\mathbb{E}[A\mathbf{x} + \mathbf{b}] = \int A\mathbf{x}f(\mathbf{x})d\mathbf{x} + \mathbf{b}f(\mathbf{x})d\mathbf{x}$$

which by integration properties we know can be written as

$$\mathbb{E}[A\mathbf{x} + \mathbf{b}] = \int A\mathbf{x}f(\mathbf{x})d\mathbf{x} + \int \mathbf{b}f(\mathbf{x})d\mathbf{x}$$

Since A and $\mathbf{b}$ are both constant we can factor them out.

$$\mathbb{E}[A\mathbf{x} + \mathbf{b}] = A \int \mathbf{x}f(\mathbf{x})d\mathbf{x} + \mathbf{b} \int f(\mathbf{x})d\mathbf{x}.$$

When rewriting this function to account for the definition, $\mathbb{E}[\mathbf{x}] = \int \mathbf{x} * f(\mathbf{x})d\mathbf{x}$, we used earlier we find that:

$$\mathbb{E}[A\mathbf{x} + \mathbf{b}] = A\mathbb{E}[\mathbf{x}] + \mathbf{b}.$$

We know that the covariance formula can be written as:

$$\text{cov}[\mathbf{x}] = \mathbf{\Sigma} = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^\top]$$

When we use our random vector $\mathbf{y} = A\mathbf{x} + \mathbf{b}$, we can rewrite this definition to be:

$$\text{cov}[\mathbf{y}] = \text{cov}[A\mathbf{x} + \mathbf{b}] = \mathbf{\Sigma} = \mathbb{E}[(A\mathbf{x} + \mathbf{b} - \mathbb{E}[A\mathbf{x} + \mathbf{b}])(A\mathbf{x} + \mathbf{b} - \mathbb{E}[A\mathbf{x} + \mathbf{b}])^\top]$$

From part a, we know that, $\mathbb{E}[A\mathbf{x} + \mathbf{b}] = A\mathbb{E}[\mathbf{x}] + \mathbf{b}$, so we can rewrite our equation as:

$$\text{cov}[A\mathbf{x} + \mathbf{b}] = \mathbb{E}[(A\mathbf{x} + \mathbf{b} - A\mathbb{E}[\mathbf{x}] - \mathbf{b})(A\mathbf{x} + \mathbf{b} - A\mathbb{E}[\mathbf{x}] - \mathbf{b})^\top]$$

This means by simplification our equation becomes:

$$\text{cov}[A\mathbf{x} + \mathbf{b}] = \mathbb{E}[(A\mathbf{x} - A\mathbb{E}[\mathbf{x}])(A\mathbf{x} - A\mathbb{E}[\mathbf{x}])^\top]$$

We can factor out A:

$$\text{cov}[A\mathbf{x} + \mathbf{b}] = \mathbb{E}[A(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^\top A^\top] = A\mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^\top]A^\top$$

Which by using the previously stated definition of covariance, we can say our equation is:

$$\text{cov}[\mathbf{y}] = \text{cov}[A\mathbf{x} + \mathbf{b}] = A\text{cov}[\mathbf{x}]A^\top = A\mathbf{\Sigma}A^\top.$$

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2 Given the dataset $\mathcal{D} = \{(x, y)\} = \{(0, 1), (2, 3), (3, 6), (4, 8)\}$

- (a) Find the least squares estimate $y = \theta^\top \mathbf{x}$ by hand using Cramer's Rule.
- (b) Use the normal equations to find the same solution and verify it is the same as part (a).
- (c) Plot the data and the optimal linear fit you found.
- (d) Find randomly generate 100 points near the line with white Gaussian noise and then compute the least squares estimate (using a computer). Verify that this new line is close to the original and plot the new dataset, the old line, and the new line.

We know that our design matrix is:

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, Y = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 8 \end{bmatrix}$$

The least squares solution satisfies the normal equations $X^T X \theta = X^T y$
We know can calculate $X^T X$.

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

By matrix multiplication we get that:

$$X^T X = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix}$$

We can calculate $X^T Y$:

$$X^T Y = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 18 \\ 56 \end{bmatrix}$$

So this means we need to solve for:

$$\begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} = \begin{bmatrix} 18 \\ 56 \end{bmatrix}$$

So we get that $4b + 9m = 18$ and $9b + 29m = 56$.

If we use Cramer's Rule, $x = D_x / D$ where D is the determinant of the coefficient matrix, we get:

$$D = \begin{vmatrix} 4 & 9 \\ 9 & 29 \end{vmatrix} = 4(29) - 9(9) = 116 - 81 = 35$$

$$D_b = \begin{vmatrix} 18 & 9 \\ 56 & 29 \end{vmatrix} = 18(29) - 9(56) = 522 - 504 = 18$$

Therefore, $b = D_b/D = 18/35$. We can calculate m the same way replacing the second column instead of the first and finding the determinant

$$D_m = \begin{vmatrix} 4 & 18 \\ 9 & 56 \end{vmatrix} = 4(56) - 18(9) = 224 - 162 = 62$$

Therefore, $m = D_m/D = 62/35$.

So our least squares line is: $y = 62/35x + 18/35$.

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